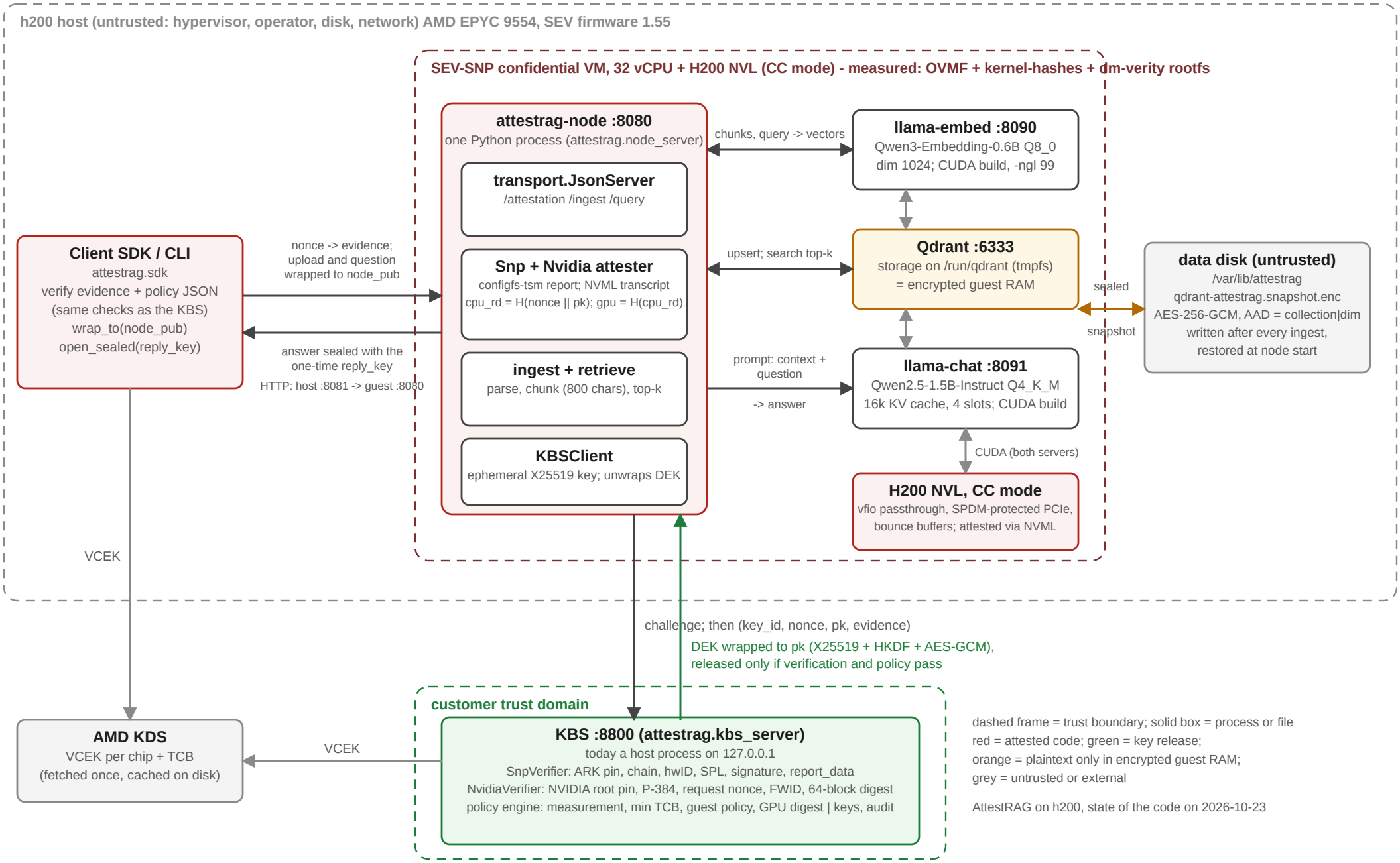
h200 host (untrusted: hypervisor, operator, disk, network) AMD EPYC 9554, SEV firmware 1.55
SEV-SNP confidential VM, 32 vCPU + H200 NVL (CC mode) - measured: OVMF + kernel-hashes + dm-verity rootfs
attestrag-node :8080
one Python process (attestrag.node_server)
transport.JsonServer
/attestation /ingest /query
Snp + Nvidia attester
configfs-tsm report; NVML transcript
cpu_rd = H(nonce || pk); gpu = H(cpu_rd)
ingest + retrieve
parse, chunk (800 chars), top-k
KBSClient
ephemeral X25519 key; unwraps DEK
chunks, query -> vectors
llama-embed :8090
Qwen3-Embedding-0.6B Q8_0
dim 1024; CUDA build, -ngl 99
upsert; search top-k
Qdrant :6333
storage on /run/qdrant (tmpfs)
= encrypted guest RAM
sealed
snapshot
data disk (untrusted)
/var/lib/attestrag
qdrant-attestrag.snapshot.enc
AES-256-GCM, AAD = collection|dim
written after every ingest,
restored at node start
prompt: context +
question
-> answer
llama-chat :8091
Qwen2.5-1.5B-Instruct Q4_K_M
16k KV cache, 4 slots; CUDA build
CUDA (both servers)
H200 NVL, CC mode
vfio passthrough, SPDM-protected PCIe,
bounce buffers; attested via NVML
Client SDK / CLI
attestrag.sdk
verify evidence + policy JSON
(same checks as the KBS)
wrap_to(node_pub)
open_sealed(reply_key)
nonce -> evidence;
upload and question
wrapped to node_pub
answer sealed with the
one-time reply_key
HTTP: host :8081 -> guest :8080
VCEK
challenge; then (key_id, nonce, pk, evidence)
DEK wrapped to pk (X25519 + HKDF + AES-GCM),
released only if verification and policy pass
customer trust domain
AMD KDS
VCEK per chip + TCB
(fetched once, cached on disk)
VCEK
KBS :8800 (attestrag.kbs_server)
today a host process on 127.0.0.1
SnpVerifier: ARK pin, chain, hwID, SPL, signature, report_data
NvidiaVerifier: NVIDIA root pin, P-384, request nonce, FWID, 64-block digest
policy engine: measurement, min TCB, guest policy, GPU digest | keys, audit
dashed frame = trust boundary; solid box = process or file
red = attested code; green = key release;
orange = plaintext only in encrypted guest RAM;
grey = untrusted or external
AttestRAG on h200, state of the code on 2026-10-23